

**EPSON OPOS ADK MANUAL**

# **APPLICATION DEVELOPMENT GUIDE**

**POSPrinter  
(TM-U120M/TM-U120IIM/TM-U120IIM-5  
8)**

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# Section 1. Introduction

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This manual describes the method of use and related items, as well as machine-specific precautions, when the EPSON TM-U120M/TM-U120IIM/TM-U120IIM-58 POS Printers are used with the EPSON OPOS ADK program.

This manual applies to the following devices.

Device List

| Serial        | Parallel       | USB | Ethernet |
|---------------|----------------|-----|----------|
| TM-U120M      | TM-U120PM      | -   | -        |
| TM-U120IIM    | TM-U120IIPM    | -   | -        |
| TM-U120IIM-58 | TM-U120IIPM-58 | -   | -        |

Before reading the manual, see the following explanation about the characteristic of the TM-U120M/TM-U120IIM/TM-U120IIM-58 models.

- Station: Receipt (9-pin Serial impact dot matrix)
- Black Mark paper

## Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

## Section 2. Details on Settings

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This section describes connection configurations and how to make the settings for the TM-U120M/TM-U120IIM/TM-U120IIM-58 models.

### 2.1. References of Firmware Versions

Refer to the release notes (Relnote.txt).

### 2.2. Port Information

#### 1) Port information when serial port is used

The port information that can be set with the SetupPOS utility is as follows.

| Item                        | Setting range         |
|-----------------------------|-----------------------|
| Baud rate [bps]             | 4800,9600,19200,38400 |
| Bit length [bit]            | 8                     |
| Parity                      | NONE, ODD, EVEN       |
| Stop bit [bit]              | 1                     |
| Handshake                   | DTR/DSR               |
| Output buffer length [byte] | 32 to 1024            |
| Output interval time [ms]   | 0 to 9999             |

The default settings are as shown in the following table.

| Item                        | Setting range |
|-----------------------------|---------------|
| Baud rate [bps]             | 9600          |
| Bit length [bit]            | 8             |
| Parity                      | NONE          |
| Stop bit [bit]              | 1             |
| Handshake                   | DTR/DSR       |
| Output buffer length [byte] | 1024          |
| Output interval time [ms]   | 2500          |

## 2) Port information when using parallel port

The port information that can be set with the SetupPOS utility is as follows.

| Item                        | Setting range |
|-----------------------------|---------------|
| Output buffer length [byte] | 32 to 1024    |
| Output interval time [ms]   | 0 to 9999     |

The default settings are as shown in the following table.

| Item                        | Setting range |
|-----------------------------|---------------|
| Output buffer length [byte] | 1024          |
| Output interval time [ms]   | 2500          |

## 2.3. Device Settings

The following explanation is about the settings for TM-U120M/TM-U120IIM/TM-U120IIM-58 models.

### 2.3.1. Usable Device Specific Settings

For the TM-U120M/TM-U120IIM/TM-U120IIM-58 models, the following device specific settings are settable by the SetupPOS utility. For the detail, please refer to the Section 2 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”.

| Tab          | Settings                                        |
|--------------|-------------------------------------------------|
| General      | Disable panel buttons                           |
|              | Assume print complete when data output finishes |
|              | Ignore firmware version check                   |
|              | Homogenize Error Codes                          |
|              | Output complete timeout [s]                     |
| Paper        | Paper Type                                      |
| Color Bitmap | Method                                          |
|              | Brightness                                      |
|              | Primary                                         |
| Status Log   | ERROR                                           |
|              | OFFLINE                                         |
|              | Log filename (full path name)                   |
|              | Maximum file size [KB]                          |

## Section 3. Function Details

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This section describes the functions of the TM-U120M/TM-U120IIM/TM-U120IIM-58 models in details. Supplementary explanation of the parts not described in detail in the "UPOS" is also given here.

### 3.1. Property Set Values and Default Values

The following explanation is about the property set values and the default values.

#### 3.1.1. Capability Set Values

The following values are the Capability set values.

| Capability Name       | Set Value     |
|-----------------------|---------------|
| CapTransaction        | TRUE          |
| CapCoverSensor        | FALSE         |
| CapConcurrentRecSlp   | FALSE         |
| CapConcurrentJrnSlp   | FALSE         |
| CapConcurrentJrnRec   | FALSE         |
| CapConcurrentPageMode | FALSE         |
| CapCharacterSet       | PTR_CCS_KANJI |
| CapMapCharacterSet    | FALSE         |
| CapJrnUnderline       | FALSE         |
| CapJrnNearEndSensor   | FALSE         |
| CapJrnItalic          | FALSE         |
| CapJrnEmptySensor     | FALSE         |
| CapJrnDwideDhigh      | FALSE         |
| CapJrnDwide           | FALSE         |
| CapJrnDhigh           | FALSE         |
| CapJrnColor           | 0             |
| CapJrnCartridgeSensor | 0             |
| CapJrnBold            | FALSE         |
| CapJrn2Color          | FALSE         |
| CapJrnPresent         | FALSE         |
| CapRecPageMode        | FALSE         |
| CapRecUnderline       | FALSE         |
| CapRecStamp           | FALSE         |
| CapRecRotate180       | FALSE         |
| CapRecRight90         | FALSE         |

| Capability Name       | Set Value                                                    |
|-----------------------|--------------------------------------------------------------|
| CapRecPapercut        | FALSE                                                        |
| CapRecNearEndSensor   | (Roll paper) TRUE<br>(Black marked roll paper) FALSE         |
| CapRecMarkFeed        | (Roll paper) 0<br>(Black marked roll paper) PTR_MF_TO_CUTTER |
| CapRecLeft90          | FALSE                                                        |
| CapRecItalic          | FALSE                                                        |
| CapRecEmptySensor     | TRUE                                                         |
| CapRecDwideDhigh      | TRUE                                                         |
| CapRecDwide           | TRUE                                                         |
| CapRecDhigh           | TRUE                                                         |
| CapRecColor           | PTR_COLOR_PRIMARY                                            |
| CapRecCartridgeSensor | 0                                                            |
| CapRecBold            | TRUE                                                         |
| CapRecBitmap          | TRUE                                                         |
| CapRecBarCode         | FALSE                                                        |
| CapRec2Color          | FALSE                                                        |
| CapRecPresent         | TRUE                                                         |
| CapSlpUnderline       | FALSE                                                        |
| CapSlpRotate180       | FALSE                                                        |
| CapSlpRight90         | FALSE                                                        |
| CapSlpNearEndSensor   | FALSE                                                        |
| CapSlpLeft90          | FALSE                                                        |
| CapSlpItalic          | FALSE                                                        |
| CapSlpEmptySensor     | FALSE                                                        |
| CapSlpDwideDhigh      | FALSE                                                        |
| CapSlpDwide           | FALSE                                                        |
| CapSlpDhigh           | FALSE                                                        |
| CapSlpColor           | 0                                                            |
| CapSlpCartridgeSensor | 0                                                            |
| CapSlpBothSidesPrint  | FALSE                                                        |
| CapSlpBold            | FALSE                                                        |
| CapSlpBitmap          | FALSE                                                        |
| CapSlpBarCode         | FALSE                                                        |
| CapSlp2Color          | FALSE                                                        |
| CapSlpFullslip        | FALSE                                                        |
| CapSlpPresent         | FALSE                                                        |
| CapSlpPageMode        | FALSE                                                        |



### 3.1.2. List Properties

The List Properties are explained in the following.

#### ● TM-U120M/TM-U120IIM

| List Property          | Settings                                           |
|------------------------|----------------------------------------------------|
| CharacterSetList       | "254,255,437,850,852,858,860,863,865,866,936,1252" |
| JrnLineCharsList       | ""                                                 |
| RecLineCharsList       | "35,42"                                            |
| SlpLineCharsList       | ""                                                 |
| RecBarCodeRotationList | ""                                                 |
| RecBitmapRotationList  | "0"                                                |
| SlpBarCodeRotationList | ""                                                 |
| SlpBitmapRotationList  | ""                                                 |
| FontTypefaceList       | ""                                                 |

#### ● TM-U120IIM-58

| List Property          | Settings                                           |
|------------------------|----------------------------------------------------|
| CharacterSetList       | "254,255,437,850,852,858,860,863,865,866,936,1252" |
| JrnLineCharsList       | ""                                                 |
| RecLineCharsList       | "25,30"                                            |
| SlpLineCharsList       | ""                                                 |
| RecBarCodeRotationList | ""                                                 |
| RecBitmapRotationList  | "0"                                                |
| SlpBarCodeRotationList | ""                                                 |
| SlpBitmapRotationList  | ""                                                 |
| FontTypefaceList       | ""                                                 |

### 3.1.3. Width and Height Properties

The width and height properties are described below.

#### ● TM-U120M/TM-U120IIM

| Property                            | Settings         |                     |                     |
|-------------------------------------|------------------|---------------------|---------------------|
|                                     | Default Value    | Maximum value [dot] | Minimum value [dot] |
| RecLineSpacing                      | 12               | 127                 | 0                   |
| JrnLineSpacing                      | X                | X                   | X                   |
| SlpLineSpacing                      | X                | X                   | X                   |
| SlpLineHeight [dot]                 | X                |                     |                     |
| RecLineHeight [dot]                 | 9                |                     |                     |
| JrnLineHeight [dot]                 | X                |                     |                     |
| SlpLineWidth [dot]                  | X                |                     |                     |
| RecLineWidth [dot]                  | 210              |                     |                     |
| JrnLineWidth [dot]                  | X                |                     |                     |
| RecSidewaysMaxLines                 | 0                |                     |                     |
| RecSidewaysMaxChars                 | 0                |                     |                     |
| RecLinesToPaperCut (No auto cutter) | 15 <sup>*1</sup> |                     |                     |
| SlpSidewaysMaxLines                 | X                |                     |                     |
| SlpSidewaysMaxChars                 | X                |                     |                     |
| SlpMaxLines                         | X                |                     |                     |

#### ● TM-U120IIM-58

| Property                            | Settings         |                     |                     |
|-------------------------------------|------------------|---------------------|---------------------|
|                                     | Default Value    | Maximum value [dot] | Minimum value [dot] |
| RecLineSpacing                      | 12               | 127                 | 0                   |
| JrnLineSpacing                      | X                | X                   | X                   |
| SlpLineSpacing                      | X                | X                   | X                   |
| SlpLineHeight [dot]                 | X                |                     |                     |
| RecLineHeight [dot]                 | 9                |                     |                     |
| JrnLineHeight [dot]                 | X                |                     |                     |
| SlpLineWidth [dot]                  | X                |                     |                     |
| RecLineWidth [dot]                  | 150              |                     |                     |
| JrnLineWidth [dot]                  | X                |                     |                     |
| RecSidewaysMaxLines                 | 0                |                     |                     |
| RecSidewaysMaxChars                 | 0                |                     |                     |
| RecLinesToPaperCut (No auto cutter) | 15 <sup>*1</sup> |                     |                     |
| SlpSidewaysMaxLines                 | X                |                     |                     |
| SlpSidewaysMaxChars                 | X                |                     |                     |
| SlpMaxLines                         | X                |                     |                     |

X : No settings

<sup>\*1</sup> It can be changed by the settings of the RecLineSpacing or the character height.

- When the RecLineSpacing is 0, the RecLinesToPaperCut is -1.

### 3.1.4. Common Property Strings

The Device information properties are described below.

#### ● TM-U120M

| I/F | DeviceName | DeviceDescription           |
|-----|------------|-----------------------------|
| S   | TM-U120M   | EPSON TM-U120M POS Printer  |
| P   | TM-U120PM  | EPSON TM-U120PM POS Printer |

#### ● TM-U120IIM

| I/F | DeviceName  | DeviceDescription             |
|-----|-------------|-------------------------------|
| S   | TM-U120IIM  | EPSON TM-U120IIM POS Printer  |
| P   | TM-U120IIPM | EPSON TM-U120IIPM POS Printer |

#### ● TM-U120IIM-58

| I/F | DeviceName     | DeviceDescription                |
|-----|----------------|----------------------------------|
| S   | TM-U120IIM-58  | EPSON TM-U120IIM-58 POS Printer  |
| P   | TM-U120IIPM-58 | EPSON TM-U120IIPM-58 POS Printer |

I/F indicates the connected interface.

The following is the list of the two connecting interfaces.

S: Serial

P: Parallel

### 3.2. Methods

The following explanation is about supported/unsupported Methods, and the detailed information.

| Method            | Supported/Unsupported                         |
|-------------------|-----------------------------------------------|
| PrintNormal       | O                                             |
| PrintTwoNormal    | X                                             |
| PrintImmediate    | O                                             |
| PrintBarCode      | X                                             |
| PrintBitmap       | O                                             |
| PrintMemoryBitmap | O                                             |
| CutPaper          | X                                             |
| MarkFeed          | (Roll paper) X<br>(Black marked roll paper) O |
| ChangePrintSide   | X                                             |
| ValidateData      | O                                             |
| TransactionPrint  | O                                             |
| SetLogo           | O                                             |
| SetBitmap         | O                                             |
| RotatePrint       | O                                             |
| EndRemoval        | X                                             |
| BeginRemoval      | X                                             |
| EndInsertion      | X                                             |
| BeginInsertion    | X                                             |
| ClearPrintArea    | X                                             |
| PageModePrint     | X                                             |

O: Supported X: Unsupported

### 3.3. Escape Sequences

The following figure is about supported/unsupported Escape Sequences.

| Escape Sequence       | Support          |
|-----------------------|------------------|
| #P                    | X                |
| #fP                   | X                |
| #sP                   | X                |
| sL                    | X                |
| #B                    | O                |
| tL                    | O                |
| bL                    | O                |
| #R                    | X                |
| #IF                   | 0~9999           |
| #uF Base Pitch [inch] | 0~ approx. 50 cm |
| #rF Maximum [inch]    | X                |
| #E                    | 0~65535          |
| #fT                   | X                |
| bC                    | O                |
| !bC                   | O                |
| #uC                   | X                |
| iC                    | X                |
| !iC                   | X                |
| #rC                   | 1                |
| rvC                   | X                |
| !rvC                  | X                |
| #sC                   | X                |
| #fC                   | X                |
| tbC                   | X                |
| !tbC                  | X                |
| tpC                   | X                |
| !tpC                  | X                |
| 1C                    | O                |
| 2C                    | O                |
| 3C                    | O                |
| 4C                    | O                |
| #hC                   | 1~2              |
| #vC                   | 1~2              |
| cA                    | X                |
| rA                    | X                |
| lA                    | X                |
| N                     | O                |

O: Supported X : Unsupported Numbers: Settable range

### 3.4. Power Condition Reports

The TM-U120M/TM-U120IIM/TM-U120IIM-58 models support Power Condition Reports as follows.

Powered on reporting : Supported

Powered off reporting : Unsupported

### 3.5. Synchronous Processing

The TM-U120M/TM-U120IIM/TM-U120IIM-58 models do not use Process ID to determine output completion.

### 3.6. Printing Positions

The TM-U120M/TM-U120IIM/TM-U120IIM-58 models support the function for setting printing position.

| Function          | Support |
|-------------------|---------|
| Left margin       | X       |
| Printing Position | X       |

O: Supported X: Unsupported

When the left margin setting function is supported, it is possible to specify the horizontal printing position of the bitmap or barcode by dots unit.

When the printing position settings are supported, it is possible to specify the horizontal printing position of the text, bitmap, or the barcode to the left, center, or the right side of the paper.

### 3.7. Electronic Logo Function (NVRAM)

The TM-U120M/TM-U120IIM/TM-U120IIM-58 models do not support the Electronic Logo Function.

### 3.8. Printable Bitmap Types and Sizes

The TM-U120M/TM-U120IIM/TM-U120IIM-58 models support the following bitmap commands. For the detail, please refer to the Section 3 of "EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)". The allowance ranges for bitmaps are as follows.

| Bitmap command type | Allowance range  |
|---------------------|------------------|
| One-line bitmap     | No setting range |

### **3.9. Maintenance Counter**

The TM-U120M/TM-U120IIM/TM-U120IIM-58 models do not support the Maintenance Counter.

### **3.10. Automatic Recovery Function**

The TM-U120M/TM-U120IIM/TM-U120IIM-58 models feature a function for automatic recovery when the power is turned on again after an interruption of power. Recovery processing is performed automatically when the printer's power is turned on again after an interruption. The recovery processing restores the printer to the condition it was in before the power was interrupted.

The other models do not support this function.

## Section 4. Warnings

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This section describes precautions in use of TM-U120M/TM-U120IIM/TM-U120IIM-58 models.

The end of the paper can not be detected when using the roll paper which is without an indicator of the end of the paper.

When the "Black Mark" roll paper is selected by SetupPOS Utility and except the "Black Mark" roll paper is set in the device, if the MarkFeed CUTTER method was executed sometimes error is returned.